

Equations of the Hierarchical Hydrogen Permeation Model

Model A: Bulk Transport Hierarchy (L1–L5) Model B: Surface-Kinetics Extension (L6)

Azeez Akinyemi

July 7, 2026

Contents

1	Introduction and notation	2
2	Model A: bulk transport hierarchy (L1–L5)	2
2.1	Summary table	2
2.2	L1: perfect metal	4
2.3	L2a: perfect oxide	5
2.4	L2b: oxide + metal in series	5
2.5	L3: defective oxide (parallel paths)	6
2.6	L4: defective metal (grain boundaries + trapping)	7
2.7	L5: full system (defective oxide + defective metal)	8
2.8	Shared temperature dependence	8
3	Model B: surface-kinetics extension (L6 and combinations)	9
3.1	Summary table	9
3.2	L6 fundamentals: Langmuir–Hinshelwood surface kinetics	10
3.3	L1+L6: perfect metal + surface kinetics	11
3.4	L2a+L6: perfect oxide + surface kinetics	11
3.5	L2+L6: oxide + metal + surface kinetics	12
3.6	L3+L6: defective oxide + surface kinetics	12
3.7	L2+L4+L6: oxide + defective metal + surface kinetics	12
3.8	L3+L4+L6: full model	13
4	Parameter values and sources	13
4.1	Bulk transport and operating conditions	13
4.2	Oxide defects (L3) and metal microstructure (L4)	14
4.3	Surface kinetics (L6)	15
5	Cross-model equation map	16
A	Resistance formulation: complete derivation	18
A.1	What a permeation resistance is	18
A.2	Oxide: linear in ΔP	19
A.3	Metal: linear in $\Delta\sqrt{P}$	19
A.4	Why the two are not directly comparable	19
A.5	Linearising the metal: origin of the factor 2	20
A.6	Consequences: regime ratio and transition pressure	20
A.7	Caveats and the Model B alternative	20

1 Introduction and notation

The permeation model is built as a hierarchy of physical fidelity. **Model A** (Section 2, notebook `Application/Proposal.ipynb`) assumes that surface processes are infinitely fast, so transport is always limited by bulk diffusion through the oxide and/or metal layers. It progresses from a perfect metal membrane (L1) through a perfect oxide (L2a), the coupled oxide–metal bilayer (L2b), a defective oxide with parallel leakage paths (L3), a defective metal with grain-boundary transport and trapping (L4), up to the full defective system (L5). **Model B** (Section 3, notebook `Application/Surface_proposal.ipynb`) relaxes the fast-surface assumption: a finite Langmuir–Hinshelwood dissociation/recombination step (L6) is placed in series with each bulk configuration, and the steady state is found by enforcing flux continuity through all layers.

All equations below are transcribed directly from the Python modules in `calculations/`; the code location column in the summary tables gives *file* $\rightarrow$ *function*. Table 1 collects the global symbols. Throughout, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ is the gas constant and T the absolute temperature in K.

Table 1: Global notation and units.

Symbol	Meaning	Units	First used
J	permeation flux	$\text{mol m}^{-2} \text{ s}^{-1}$	Eq. (2)
C	dissolved hydrogen concentration	mol m^{-3}	Eq. (1)
$P_{\text{up}}, P_{\text{down}}$	upstream / downstream H_2 pressure	Pa	Eq. (1)
P_{int}	oxide–metal interface pressure	Pa	Eq. (8)
$D_{\text{m}}, D_{\text{ox}}$	diffusivity of H in metal / H_2 in oxide	$\text{m}^2 \text{ s}^{-1}$	Eq. (2)
K_{s}	Sieverts’ solubility constant (metal)	$\text{mol m}^{-3} \text{ Pa}^{-1/2}$	Eq. (1)
K_{ox}	oxide solubility constant (Henry form in A, Sieverts form in B; §3.2)	$\text{mol m}^{-3} \text{ Pa}^{-1}$ (A) $\text{mol m}^{-3} \text{ Pa}^{-1/2}$ (B)	Eq. (5)
$L_{\text{m}}, L_{\text{ox}}$	metal / oxide thickness	m	Eq. (3)
Φ	metal permeability $D_{\text{m}}K_{\text{s}}$	$\text{mol m}^{-1} \text{ s}^{-1} \text{ Pa}^{-1/2}$	Eq. (4)
D_{eff}	effective (microstructure-modified) diffusivity	$\text{m}^2 \text{ s}^{-1}$	Eq. (21)
θ	surface coverage (fraction of occupied sites)	–	Eq. (28)
k_{diss}	dissociative adsorption rate constant	$\text{mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$	Eq. (28)
K_{eq}	surface equilibrium constant $k_{\text{diss}}/k_{\text{recomb}}$	Pa^{-1}	Eq. (29)
α	oxide permeance $D_{\text{ox}}K_{\text{ox}}/L_{\text{ox}}$ (Model B)	$\text{mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1/2}$	Eq. (35)
β	metal permeance $D_{\text{m}}K_{\text{s}}/L_{\text{m}}$	$\text{mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1/2}$	Eq. (35)
$E_{\text{D}}, H_{\text{s}}, E$	activation / solution energies	J mol^{-1}	Eq. (26)

2 Model A: bulk transport hierarchy (L1–L5)

2.1 Summary table

Table 2: Model A equations, code locations (file, function in `calculations/`), and references.

Eq.	Name	Formula	Code location	Ref.
<i>L1 — perfect metal</i>				
(1)	Sieverts’ law	$C = K_{\text{s}}\sqrt{P}$	<code>permeation.calc.py</code> <code>sieverts_concentration</code>	Sieverts (1929)
(2)	Fick’s first law	$J = D(C_{\text{up}} - C_{\text{down}})/L$	<code>permeation.calc.py</code> <code>fick_flux</code>	Fick (1855)
(3)	L1 flux	$J_{\text{L1}} = \frac{D_{\text{m}}K_{\text{s}}}{L_{\text{m}}}(\sqrt{P_{\text{up}}} - \sqrt{P_{\text{down}}})$	<code>permeation.calc.py</code> <code>calculate_simple_metal_flux</code>	

Eq.	Name	Formula	Code location	Ref.
(4)	Permeability	$\Phi = D_m K_s$	utils.py get_permeability	
<i>L2a — perfect oxide</i>				
(5)	Henry's law	$C = K_{ox} P$	oxide_permeation.py molecular_diffusion_flux	
(6)	L2a flux	$J_{L2a} = \frac{D_{ox} K_{ox}}{L_{ox}} (P_{up} - P_{down})$	oxide_permeation.py molecular_diffusion_flux	
(7)	Oxide resistance	$R_{oxide} = L_{ox} / (D_{ox} K_{ox})$	oxide_permeation.py calculate_oxide_resistance	
<i>L2b — oxide + metal bilayer</i>				
(8)	Flux continuity	$J_{ox}(P_{int}) = J_m(P_{int})$	interface_solver.py flux_balance_equation	
(9)	Linearised metal resistance	$R_{metal} \approx 2L_m \sqrt{P_{int}} / (D_m K_s)$	oxide_permeation.py calculate_metal_resistance	
(10)	Regime ratio	$\chi = R_{oxide} / R_{metal}$	interface_solver.py calculate_oxide_metal_system	
(11)	Transition pressure	$P_{trans} = \left[\frac{L_{ox} D_m K_s}{2 D_{ox} K_{ox} L_m} \right]^2$	oxide_permeation.py calculate_transition_pressure	
<i>L3 — defective oxide (parallel paths)</i>				
(12)	Parallel-path flux	$J_{L3} = f_{int} J_{int} + f_{def} J_{def}$	parallel_oxide_defect_paths.py calculate_parallel_path_flux	Strehlow and Savage (1974); Zarchy and Axtmann (1979)
(13)	Crack path	$L_{ox}^{crack} = \gamma L_{ox}$	parallel_oxide_defect_paths.py calculate_defect_path_flux	Strehlow and Savage (1974)
(14)	Oxide-GB path	$D_{ox}^{gb} = \delta_D D_{ox}$	parallel_oxide_defect_paths.py calculate_defect_path_flux	
(15)	Mixed defects	$J_{def} = \sum_i w_i J_i$	parallel_oxide_defect_paths.py calculate_defect_path_flux	
<i>L4 — defective metal (grain boundaries + traps)</i>				
(16)	GB volume fraction	$f_{gb} = 3\delta/d$	defective_metal.py grain_boundary_density	Underwood (1970); Palumbo and Aust (1990)
(17)	GB enhancement	$\alpha_{gb}(T) = A \exp\left[\frac{Q_{bulk} - Q_{gb}}{RT}\right]$	defective_metal.py gb_enhancement_factor	Tsuru and Latanision (1982); Brass and Chanfreau (1996)
(18)	GB parallel model	$D_{gb-enh} = (1 - f_{gb})D + f_{gb} \alpha_{gb} D$	defective_metal.py calculate_gb_enhanced_diffusivity	
(19)	Hart equation	see text	defective_metal.py calculate_gb_enhanced_diffusivity	Hart (1957)
(20)	Trap occupancy	$\theta_T = \frac{K \theta_L}{1 + K \theta_L}$	defective_metal.py trap_occupancy	Oriani (1970); McNabb and Foster (1963)
(21)	Trapping reduction	$D_{eff} = \frac{D}{1 + \sum_i N_{T,i} K_i / N_L}$	defective_metal.py calculate_effective_diffusivity_trapping	Oriani (1970)
(22)	Combined model	$D_{eff} = \frac{(1 - f_{gb})D + f_{gb} \alpha_{gb} D}{1 + \sum_i N_{T,i} K_i / N_L}$	defective_metal.py combined_microstructure_model	Oudriss et al. (2012)
(23)	Vacancy concentration	$C_v = N \exp(-E_f^v / RT)$	defective_metal.py vacancy_concentration	Kraftmakher (1998)
(24)	L4 flux	$J_{L4} = \frac{D_{eff} K_s}{L_m} (\sqrt{P_{up}} - \sqrt{P_{down}})$	permeation_calc.py calculate_defective_metal_flux	
<i>L5 — full system</i>				

Eq.	Name	Formula	Code location	Ref.
(25)	Coupled closure	$J_{\text{ox}}(P_{\text{int}}) = J_{\text{L4}}(P_{\text{int}}; D_{\text{eff}})$	<code>interface_solver.py</code> <code>solve_interface_pressure_defective_metal</code>	
<i>Shared temperature dependence</i>				
(26)	Arrhenius (reference form)	$k(T) = k_{\text{ref}} \exp\left[-\frac{E}{R}\left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right]$	<code>utils.py</code> <code>arrhenius</code>	
(27)	Permeation activation	$Q_{\Phi} = E_D + H_s$	<code>utils.py</code> <code>get_permeability</code>	

2.2 L1: perfect metal

Sieverts’ law. At a clean metal surface in equilibrium with H_2 gas, the molecule dissociates and dissolves atomically, so the dissolved concentration scales with the *square root* of pressure:

$$C = K_s \sqrt{P}. \quad (1)$$

Here C is the hydrogen concentration at the surface [mol m^{-3}], P the hydrogen partial pressure [Pa], and K_s the Sieverts solubility constant [$\text{mol m}^{-3} \text{Pa}^{-1/2}$], a material property with its own van ’t Hoff temperature dependence (Eq. (26) with $E = H_s$). The square root is the fingerprint of dissociative absorption (Sieverts, 1929): it is what distinguishes metals from oxides in every log–log flux plot in the model.

Fick’s first law. Steady-state diffusion through a flat membrane with a linear concentration profile gives

$$J = D \frac{C_{\text{up}} - C_{\text{down}}}{L}, \quad (2)$$

where J is the flux [$\text{mol m}^{-2} \text{s}^{-1}$], D the diffusion coefficient [$\text{m}^2 \text{s}^{-1}$], L the membrane thickness [m], and C_{up} , C_{down} the boundary concentrations [mol m^{-3}]. Positive flux runs from upstream to downstream (Fick, 1855).

L1 permeation flux. Substituting Eq. (1) at both faces into Eq. (2) yields the classical Richardson form,

$$J_{\text{L1}} = \frac{D_{\text{m}} K_s}{L_{\text{m}}} \left(\sqrt{P_{\text{up}}} - \sqrt{P_{\text{down}}} \right), \quad (3)$$

with D_{m} the metal diffusivity [$\text{m}^2 \text{s}^{-1}$], L_{m} the metal thickness [m], and P_{up} , P_{down} the boundary pressures [Pa] (Richardson et al., 1904; Sieverts, 1929). On a log–log plot of J versus P_{up} (with $P_{\text{down}} \approx 0$) this gives the characteristic slope of 0.5; the notebooks use this slope as the L1 validation criterion.

Permeability. The kinetic and thermodynamic factors are lumped into

$$\Phi = D_{\text{m}} K_s, \quad (4)$$

with units $\text{mol m}^{-1} \text{s}^{-1} \text{Pa}^{-1/2}$. Written with Φ , Eq. (3) becomes $J = (\Phi/L_{\text{m}}) \Delta\sqrt{P}$, and the corresponding driving-force resistance is $R_{\text{L1}} = L_{\text{m}}/\Phi$ [$\text{Pa}^{1/2} \text{s m}^2 \text{mol}^{-1}$]. Note this resistance acts on a $\sqrt{P}$ driving force, so it is *not* directly comparable to the oxide resistance of Eq. (7) without linearisation (Eq. (9)).

2.3 L2a: perfect oxide

Henry’s law. In the oxide, hydrogen is assumed to diffuse as intact H_2 molecules (no dissociation), so solubility is *linear* in pressure:

$$C = K_{\text{ox}} P, \quad (5)$$

where K_{ox} is the Henry constant [$\text{mol m}^{-3} \text{Pa}^{-1}$] — note the different units from K_{s} . This single change of exponent propagates through the whole hierarchy: it is why oxide-limited and metal-limited regimes are distinguishable by pressure slope.

L2a flux. Combining Eq. (5) with Fick’s law across the oxide thickness L_{ox} :

$$J_{\text{L2a}} = \frac{D_{\text{ox}} K_{\text{ox}}}{L_{\text{ox}}} (P_{\text{up}} - P_{\text{down}}), \quad (6)$$

with D_{ox} the molecular diffusivity of H_2 in the oxide [$\text{m}^2 \text{s}^{-1}$] (Strehlow and Savage, 1974; Hollenberg et al., 1995). The flux is linear in ΔP (log–log slope 1.0), the L2a validation criterion. The Cr_2O_3 transport parameters themselves come from Nemanić et al. (2023) (Φ_{ox} , D_{ox}), Stöver et al. (1986) ($Q_{p,\text{ox}}$), and Chen et al. (2011) ($E_{D,\text{ox}}$ from DFT).

Oxide resistance. Because transport is linear, a genuine pressure-independent resistance exists:

$$R_{\text{oxide}} = \frac{L_{\text{ox}}}{D_{\text{ox}} K_{\text{ox}}}, \quad J = \frac{\Delta P}{R_{\text{oxide}}}, \quad (7)$$

with R_{oxide} in $\text{Pa s m}^2 \text{mol}^{-1}$. This constancy is the key structural difference from the metal layer and is what makes the series combination in L2b a nonlinear problem.

2.4 L2b: oxide + metal in series

Flux continuity closure. With an oxide on the upstream face of the metal, an unknown interface pressure P_{int} develops at the oxide–metal boundary. At steady state the same flux must cross both layers:

$$\underbrace{\frac{D_{\text{ox}} K_{\text{ox}}}{L_{\text{ox}}} (P_{\text{up}} - P_{\text{int}})}_{J_{\text{ox}}(P_{\text{int}})} = \underbrace{\frac{D_{\text{m}} K_{\text{s}}}{L_{\text{m}}} (\sqrt{P_{\text{int}}} - \sqrt{P_{\text{down}}})}_{J_{\text{m}}(P_{\text{int}})}. \quad (8)$$

This series-barrier construction is the standard treatment of oxide permeation barriers on structural metals (Strehlow and Savage, 1974; Hollenberg et al., 1995). The mixed linear/square-root equation has no closed-form solution; the code brackets $P_{\text{int}} \in (P_{\text{down}}, P_{\text{up}})$ and solves it with Brent’s method (`interface_solver.py` → `solve_interface_pressure`). The solved P_{int} determines both the flux and the operating regime.

Linearised metal resistance. To compare the two layers on a common ΔP basis, the metal flux is linearised about P_{int} using $\partial\sqrt{P}/\partial P = 1/(2\sqrt{P})$:

$$R_{\text{metal}} \approx \frac{2L_{\text{m}}\sqrt{P_{\text{int}}}}{D_{\text{m}}K_{\text{s}}}, \quad (9)$$

in $\text{Pa s m}^2 \text{mol}^{-1}$. Unlike R_{oxide} , this resistance grows with $\sqrt{P_{\text{int}}}$: at high pressure the metal becomes comparatively easier to traverse per pascal of driving force.

Regime classification. The resistance ratio

$$\chi = \frac{R_{\text{oxide}}}{R_{\text{metal}}} \begin{cases} \chi > 10 & \text{oxide-limited } (P_{\text{int}} \ll P_{\text{up}}, \text{ slope} \rightarrow 1.0), \\ 0.5 \leq \chi \leq 10 & \text{transition } (0.5 < \text{slope} < 1.0), \\ \chi < 0.5 & \text{metal-limited } (P_{\text{int}} \rightarrow P_{\text{up}}, \text{ slope} \rightarrow 0.5), \end{cases} \quad (10)$$

is evaluated at the solved interface pressure. Setting $R_{\text{oxide}} = R_{\text{metal}}$ and solving for pressure gives the approximate crossover

$$P_{\text{trans}} = \left(\frac{L_{\text{ox}} D_{\text{m}} K_{\text{s}}}{2 D_{\text{ox}} K_{\text{ox}} L_{\text{m}}} \right)^2, \quad (11)$$

the pressure at which the rate-limiting layer switches — a useful design number for choosing operating windows.

Remark (comparing metal and oxide resistances — the chemistry behind the units).

The mismatch between the two resistances is not a mathematical artefact; it records *which chemical species each layer transports*. The oxide carries *intact* H_2 : no bond is broken on dissolution, one gas molecule becomes one dissolved molecule, so $C \propto P$ (Henry) and the flux responds linearly to ΔP with the constant resistance R_{oxide} [$\text{Pa s m}^2 \text{ mol}^{-1}$]. Dissociation is deferred to the oxide–metal interface, where $\text{H}_2 \rightleftharpoons 2\text{H}^*$; the law of mass action for a reaction delivering *two* atoms per molecule gives $C_{\text{H}}^2 \propto P_{\text{int}}$, i.e. $C_{\text{H}} \propto \sqrt{P_{\text{int}}}$ — Sieverts’ law *is* that 2:1 stoichiometry. The metal therefore responds to $\Delta\sqrt{P}$ (natural resistance $R_{\text{L1}} = L_{\text{m}}/\Phi$ [$\text{Pa}^{1/2} \text{ s m}^2 \text{ mol}^{-1}$]): doubling the pressure less than doubles the dissolved-atom population, so per pascal the metal becomes progressively harder to feed while the oxide does not. Eq. (9) translates the metal onto the oxide’s per-pascal basis by linearising about P_{int} (the factor 2 is $d\sqrt{P}/dP = 1/(2\sqrt{P})$, the marginal yield of the dissociation equilibrium), which is what makes χ in Eq. (10) meaningful. Two caveats: the linearised R_{metal} is a *diagnostic only* — the L2b flux is always computed from the exact continuity condition Eq. (8), never as $\Delta P/(R_{\text{oxide}} + R_{\text{metal}})$ — and Model B never meets the mismatch because there dissociation happens at the *gas–oxide* surface instead, so atomic H traverses every layer and all driving forces are already in $\sqrt{P}$ units (§3.2). Appendix A gives the complete derivation.

2.5 L3: defective oxide (parallel paths)

Area-weighted parallel paths. Following [Strehlow and Savage \(1974\)](#), a real oxide is treated as intact film and defects acting as parallel permeation paths, weighted by their area fractions:

$$J_{\text{L3}} = f_{\text{intact}} J_{\text{intact}} + f_{\text{defect}} J_{\text{defect}}, \quad f_{\text{intact}} + f_{\text{defect}} = 1, \quad (12)$$

where f_{defect} is the fraction of surface area occupied by defects [–] and each J is a flux *density* through its own path [$\text{mol m}^{-2} \text{ s}^{-1}$]. Crucially, the intact oxide still conducts (J_{intact} from the full L2b solver), so even $\sim 1\%$ defect coverage can dominate without the intact film being ignored ([Zarchy and Axtmann, 1979](#)). Each defect type maps onto a modified L1 or L2b problem:

$$\text{pinhole: } J_{\text{def}} = J_{\text{L1}} \quad (\text{bare metal, Eq. (3); no oxide barrier}),$$

$$\text{crack: } L_{\text{ox}}^{\text{crack}} = \gamma L_{\text{ox}}, \quad 0 < \gamma < 1 \quad (\text{default } \gamma = 0.1), \quad (13)$$

$$\text{oxide grain boundary: } D_{\text{ox}}^{\text{gb}} = \delta_D D_{\text{ox}}, \quad \delta_D > 1 \quad (\text{default } \delta_D = 10), \quad (14)$$

where γ is the crack thickness factor [–] and δ_D the grain-boundary diffusivity enhancement [–]; crack and GB paths are solved with the L2b machinery using the modified oxide properties. Several defect types can coexist; their contribution is the normalised weighted sum

$$J_{\text{defect}} = \frac{\sum_i w_i J_i}{\sum_i w_i}, \quad (15)$$

with w_i the component area fractions within the defective region.

2.6 L4: defective metal (grain boundaries + trapping)

Grain-boundary volume fraction. From stereology of a random polycrystal (Underwood, 1970), the GB surface area per volume is $S_v = s/d$ and the GB volume fraction is

$$f_{\text{gb}} = \frac{s \delta}{d}, \quad s = \begin{cases} 3 & \text{equiaxed grains} \\ 2 & \text{columnar} \\ 1 & \text{planar/lamellar,} \end{cases} \quad (16)$$

where δ is the GB thickness [m] (default 0.5 nm), d the mean grain diameter [m], and s the dimensionality shape factor [-]. The same geometry supplies the GB *trap* density $N_{\text{gb}} = S_v \rho_{\text{gb}}$ [m^{-3}], with $\rho_{\text{gb}} \approx 10^{19} \text{ m}^{-2}$ the areal density of trap sites on the boundary.

Grain-boundary enhancement factor. GBs are fast diffusion paths because their activation energy is lower ($Q_{\text{gb}} \approx 0.6 Q_{\text{bulk}}$):

$$\alpha_{\text{gb}}(T) = \frac{D_{\text{gb}}}{D_{\text{bulk}}} = A \exp \left[\frac{Q_{\text{bulk}} - Q_{\text{gb}}}{RT} \right], \quad (17)$$

where A is the pre-exponential ratio $D_{0,\text{gb}}/D_{0,\text{bulk}}$ [-] and Q are activation energies [J mol^{-1}] (Tsuru and Latanision, 1982; Brass and Chanfreau, 1996). In the code the factor is interpolated log-linearly in $1/T$ from tabulated austenitic-steel data ($\alpha_{\text{gb}} \approx 500$ at 600°C down to ≈ 20 at 1000°C), then scaled by GB character (HAGB 1.0, LAGB 0.1, twin 0.05, special 0.3).

GB-enhanced diffusivity. Treating lattice and GB as independent parallel channels:

$$D_{\text{gb-enh}} = (1 - f_{\text{gb}}) D_{\text{bulk}} + f_{\text{gb}} \alpha_{\text{gb}} D_{\text{bulk}} = D_{\text{bulk}} [1 + f_{\text{gb}}(\alpha_{\text{gb}} - 1)]. \quad (18)$$

For larger GB fractions ($f_{\text{gb}} \gtrsim 0.01$), where network connectivity matters, the Hart equation (Hart, 1957) is available instead:

$$\frac{D_{\text{eff}}}{D_{\text{bulk}}} = 1 + \frac{\frac{f_{\text{gb}}}{1 - f_{\text{gb}}} (\alpha_{\text{gb}} - 1)}{1 + \frac{2(\alpha_{\text{gb}} - 1) f_{\text{gb}}}{3(1 - f_{\text{gb}})}}. \quad (19)$$

Oriani trap occupancy. Traps (dislocations, vacancies, GBs, precipitates) immobilise hydrogen. Under local equilibrium (Oriani, 1970), each trap type i has an equilibrium constant and occupancy

$$K_i = \exp \left(\frac{E_{b,i}}{RT} \right), \quad \theta_{T,i} = \frac{K_i \theta_L}{1 + K_i \theta_L}, \quad \theta_L = \frac{N_A C_L}{N_L}, \quad (20)$$

where $E_{b,i}$ is the trap binding energy [J mol^{-1}] (dislocations 20–30, GBs 40–50, vacancies 40–45, precipitates 60–90 kJ mol^{-1}), θ_L the lattice-site occupancy [-], C_L the mobile (lattice) hydrogen concentration [mol m^{-3}], N_L the interstitial site density [m^{-3}] ($\approx 1.06 \times 10^{29}$ for FCC), and N_A Avogadro's number. The model is valid for $\theta_{T,i} < 0.9$; beyond that the kinetic McNabb–Foster treatment (McNabb and Foster, 1963) is recommended.

Trapping-reduced diffusivity. Partitioning hydrogen between mobile and trapped populations gives the Oriani reduction

$$D_{\text{eff}} = \frac{D}{1 + \sum_i \frac{N_{T,i} K_i}{N_L}}, \quad (21)$$

where $N_{T,i}$ is the trap density of type i [m^{-3}]. The denominator term $\sum_i N_{T,i} K_i / N_L$ is the ratio of trapped to mobile hydrogen; the critical trap density at which $D_{\text{eff}} = D/2$ is $N_T^* = N_L / K$.

Combined microstructure model. Enhancement and trapping are applied sequentially (GB first, then traps, assumed uniformly distributed):

$$D_{\text{eff}} = \frac{(1 - f_{\text{gb}}) D_{\text{bulk}} + f_{\text{gb}} \alpha_{\text{gb}} D_{\text{bulk}}}{1 + \sum_i \frac{N_{T,i} K_i}{N_L}}. \quad (22)$$

The net modification factor $M = D_{\text{eff}}/D_{\text{bulk}}$ can be either > 1 (GB-dominated, typically coarse traps / high T) or < 1 (trapping-dominated, typically fine microstructure / low T) (Oudriss et al., 2012). Thermal vacancies, one of the trap populations, follow

$$C_v = N \exp\left(-\frac{E_f^v}{RT}\right), \quad (23)$$

with E_f^v the vacancy formation energy [J mol^{-1}] (140 kJ mol^{-1} for Incoloy 800) and N the lattice site density [m^{-3}] (Kraftmakher, 1998).

L4 permeation flux. The membrane flux keeps the Sieverts boundary conditions (solubility is a bulk thermodynamic property, unchanged by microstructure) but replaces D with D_{eff} :

$$J_{\text{L4}} = \frac{D_{\text{eff}} K_s}{L_m} \left(\sqrt{P_{\text{up}}} - \sqrt{P_{\text{down}}} \right). \quad (24)$$

Because trap occupancy depends on the local concentration $C(x)$ — the profile linear in $\sqrt{P}$ -space, written out as Eq. (45) — D_{eff} varies through the thickness; the code evaluates it on an n -point profile and averages (arithmetic, harmonic, inlet, or outlet — the `method` argument) before applying Eq. (24).

2.7 L5: full system (defective oxide + defective metal)

L5 has no separate physics: it composes the L3 oxide description with the L4 metal description through the same flux-continuity closure as L2b, now with a concentration-dependent D_{eff} :

$$\frac{D_{\text{ox}} K_{\text{ox}}}{L_{\text{ox}}} (P_{\text{up}} - P_{\text{int}}) = \frac{D_{\text{eff}}(P_{\text{int}}) K_s}{L_m} (\sqrt{P_{\text{int}}} - \sqrt{P_{\text{down}}}). \quad (25)$$

Since D_{eff} depends on the concentration profile, which depends on P_{int} , the solver iterates: guess P_{int} , evaluate D_{eff} on the resulting profile, re-solve Eq. (25) by Brent’s method, and repeat until P_{int} converges (`interface_solver.py` $\rightarrow$ `solve_interface_pressure_defective_metal`). The defective-oxide variant applies this closure per parallel path of Eq. (12); the assembly is orchestrated in `Application/Proposal.ipynb`. Regime classification reuses Eq. (10) with R_{metal} evaluated at D_{eff} .

2.8 Shared temperature dependence

Every transport property in both models (D_m , K_s , D_{ox} , K_{ox} , k_{diss} , K_{eq}) is evaluated at temperature with the same *reference-temperature* Arrhenius form:

$$k(T) = k_{\text{ref}} \exp\left[-\frac{E}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right], \quad (26)$$

where k_{ref} is the measured value at the reference temperature T_{ref} [K] and E the relevant activation or solution energy [J mol^{-1}]: E_D for diffusivities, H_s (metal) or $H_{\text{sol,ox}}$ (oxide) for solubilities, E_{diss} for the dissociation rate constant. This form avoids extrapolating pre-exponentials far outside the fitted range. The permeability inherits

$$Q_\Phi = E_D + H_s, \quad \Phi(T) = \Phi_{\text{ref}} \exp\left[-\frac{Q_\Phi}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right], \quad (27)$$

so an Arrhenius plot of measured permeation flux yields Q_Φ directly — the basis of the Arrhenius validation panels in both notebooks. Material parameters (D_{ref} , E_D , $K_{s,\text{ref}}$, H_s , oxide analogues, surface kinetics) live in `calculations/config/model_config.py` and `data/`; Section 4 lists every value with its default, sensitivity range, and literature source (Schmidt and Batfalsky, 1985; Nemanič et al., 2023; Stöver et al., 1986; Chen et al., 2011; Grant et al., 1988; Zhu et al., 2021; Lu et al., 2022; Young and Scully, 1997).

3 Model B: surface-kinetics extension (L6 and combinations)

Model B inserts a finite dissociation/recombination step at the gas–solid interface, in series with the Model A bulk layers. All of Model B lives in `calculations/surface_kinetics.py`; the bulk equations (1)–(27) are reused unchanged.

3.1 Summary table

Table 3: Model B equations, code locations, and references. All code locations are functions in `calculations/surface_kinetics.py` unless noted.

Eq.	Name	Formula	Code location	Ref.
<i>L6 fundamentals</i>				
(28)	Dissociative adsorption	$r_{\text{ads}} = k_{\text{diss}} P(1 - \theta)^2$	<code>surface_flux</code>	Pick and Sonnenberg (1985); Baskes (1980)
(29)	Recombinative desorption	$r_{\text{des}} = (k_{\text{diss}}/K_{\text{eq}}) \theta^2$	<code>surface_flux</code>	Pick and Sonnenberg (1985); Wampler (1986)
(30)	Net surface flux	$J_{\text{surf}} = k_{\text{diss}} [P(1 - \theta)^2 - \theta^2/K_{\text{eq}}]$	<code>surface_flux</code>	Andrew and Haasz (1992)
(31)	Langmuir isotherm	$\theta_{\text{eq}} = \frac{\sqrt{K_{\text{eq}} P}}{1 + \sqrt{K_{\text{eq}} P}}$	implicit ($J_{\text{surf}} = 0$)	Langmuir (1918)
(32)	Coverage conversion	$g(\theta) = \frac{\theta}{(1 - \theta)\sqrt{K_{\text{eq}}}}$	<code>g_theta</code>	
(34)	Surface resistance	$R_{\text{surface}} \approx \frac{1}{k_{\text{diss}} P_{\text{up}}(1 - \theta)^2}$	<code>smooth_surface_resistance</code>	
(35)	Permeances	$\alpha = \frac{D_{\text{ox}} K_{\text{ox}}}{L_{\text{ox}}}, \beta = \frac{D_{\text{m}} K_{\text{s}}}{L_{\text{m}}}$	<code>compute_permeances</code>	
(33)	Sieverts–Langmuir link	$K_{\text{s}} = \sqrt{k_{\text{diss}}/k_{\text{recomb}}} N_{\text{surf}}/N_{\text{L}}$	<code>data/surface_kinetics_data.py</code>	Pick and Sonnenberg (1985)
<i>L1+L6 — metal + surface</i>				
(36)	Metal flux	$J_{\text{m}} = \beta(g(\theta) - \sqrt{P_{\text{down}}})$	<code>metal_flux_L1</code>	
(37)	Closure in θ	$J_{\text{surf}}(\theta) = J_{\text{m}}(\theta)$	<code>solve_steady_state_flux_L1L6</code>	
<i>L2a+L6 — oxide + surface</i>				
(38)	Oxide flux	$J_{\text{ox}} = \alpha(g(\theta) - \sqrt{P_{\text{down}}})$	<code>oxide_flux_L2a</code>	
<i>L2+L6 — oxide + metal + surface</i>				
(39)	Analytic interface pressure	$\sqrt{P_{\text{int}}} = \frac{\alpha g(\theta) + \beta \sqrt{P_{\text{down}}}}{\alpha + \beta}$	<code>sqrt_P_int_from_theta</code>	
(40)	Three-flux system	$J_{\text{surf}} = J_{\text{ox}} = J_{\text{m}}$	<code>solve_steady_state_flux</code>	
(41)	Resistance fractions	$f_i = R_i/R_{\text{total}}$	<code>solve_steady_state_flux</code>	
<i>L3+L6 — defective oxide + surface</i>				

Eq.	Name	Formula	Code location	Ref.
(42)	Path permeances	$\alpha_{\text{crack}} = \alpha/\gamma, \alpha_{\text{gb}} = \delta_D \alpha, \alpha_{\text{pin}} = \infty$	<code>calculate_parallel_path_flux_L6</code>	Strehlow and Savage (1974)
(43)	Multi-path total	$J_{\text{tot}} = f_{\text{int}} J_{\text{int}} + \sum_i f_i J_i$	<code>calculate_mixed_defect_flux_L6</code>	
<i>L2+L4+L6 — oxide + defective metal + surface</i>				
(44)	Effective permeance	$\beta_{\text{eff}} = D_{\text{eff}} K_{\text{s}}/L_{\text{m}}$	<code>calculate_defective_metal_flux_L6</code>	
(45)	Concentration profile	$C(x) = K_{\text{s}} [\sqrt{P_{\text{int}}} - (\sqrt{P_{\text{int}}} - \sqrt{P_{\text{down}}}) \frac{x}{L_{\text{m}}}]$	<code>calculate_defective_metal_flux_L6</code>	
<i>L3+L4+L6 — full model</i>				
(46)	Full-model total	$J_{\text{tot}} = f_{\text{int}} J_{\text{int}} + \sum_i f_i J_i$	<code>calculate_full_model_flux_L346_v2</code>	

3.2 L6 fundamentals: Langmuir–Hinshelwood surface kinetics

Dissociative adsorption. An H_2 molecule needs two adjacent vacant surface sites to dissociate, $\text{H}_2(\text{g}) + 2S \rightarrow 2\text{H}(\text{ads})$, giving a rate second-order in the vacancy fraction:

$$r_{\text{ads}} = k_{\text{diss}} P (1 - \theta)^2, \quad (28)$$

where $\theta \in [0, 1]$ is the surface coverage (fraction of occupied sites) $[-]$, k_{diss} the dissociation rate constant $[\text{mol m}^{-2} \text{s}^{-1} \text{Pa}^{-1}]$, and P the local H_2 pressure $[\text{Pa}]$ (Pick and Sonnenberg, 1985; Baskes, 1980). k_{diss} follows Eq. (26) with activation energy E_{diss} ; the numerical rate constants for both the oxide surface and the clean metal surface (used for pinhole paths) are taken from the 316-steel measurements of Grant et al. (1988), the only literature source providing the full dissociation constant k_1 under clean, partially cleaned, and oxidised surface conditions.

Recombinative desorption. The reverse step, $2\text{H}(\text{ads}) \rightarrow \text{H}_2(\text{g}) + 2S$, is second-order in coverage. The code parameterises the recombination constant through the surface equilibrium constant $K_{\text{eq}} = k_{\text{diss}}/k_{\text{recomb}}$ $[\text{Pa}^{-1}]$:

$$r_{\text{des}} = k_{\text{recomb}} \theta^2 = \frac{k_{\text{diss}}}{K_{\text{eq}}} \theta^2. \quad (29)$$

Net surface flux. The net rate of atoms entering the solid is the Langmuir–Hinshelwood balance

$$J_{\text{surf}}(\theta) = k_{\text{diss}} \left[P_{\text{up}} (1 - \theta)^2 - \frac{\theta^2}{K_{\text{eq}}} \right], \quad (30)$$

positive for net adsorption (Andrew and Haasz, 1992). This is the extra series element that Model B adds to every bulk configuration.

Langmuir isotherm. Setting $J_{\text{surf}} = 0$ recovers the equilibrium coverage

$$\frac{\theta_{\text{eq}}}{1 - \theta_{\text{eq}}} = \sqrt{K_{\text{eq}} P} \implies \theta_{\text{eq}} = \frac{\sqrt{K_{\text{eq}} P}}{1 + \sqrt{K_{\text{eq}} P}}, \quad (31)$$

which grows as $\sqrt{P}$ at low pressure and saturates at $\theta_{\text{eq}} \rightarrow 1$ (Langmuir, 1918). When the surface is fast ($\theta \rightarrow \theta_{\text{eq}}$) Model B collapses onto Model A — this limit is used as a verification test in `Application/Surface_proposal.ipynb`.

Coverage-to-driving-force conversion. Inverting Eq. (31) defines the effective $\sqrt{P}$ that a coverage θ presents to the bulk layers beneath:

$$g(\theta) = \frac{\theta}{(1 - \theta)\sqrt{K_{\text{eq}}}} \quad [\text{Pa}^{1/2}]. \quad (32)$$

$g(\theta)$ is the pivot of the whole Model B formulation: it lets every bulk flux be written as a function of the single unknown θ , so the coupled surface–bulk problem reduces to one-dimensional root finding. It also connects to the Sieverts constant: consistency of the surface equilibrium with bulk solubility requires

$$K_s = \sqrt{k_{\text{diss}}/k_{\text{recomb}}} \frac{N_{\text{surf}}}{N_L}, \quad (33)$$

with $N_{\text{surf}} \approx 10^{19} \text{ m}^{-2}$ the surface site density (Pick and Sonnenberg, 1985).

Surface resistance and permeances. Linearising Eq. (30) gives the surface resistance used for regime classification,

$$R_{\text{surface}} \approx \frac{1}{k_{\text{diss}} P_{\text{up}} (1 - \theta)^2} \quad [\text{Pa}^{1/2} \text{ s m}^2 \text{ mol}^{-1}], \quad (34)$$

which blows up as the surface saturates ($\theta \rightarrow 1$) or as $k_{\text{diss}} P_{\text{up}}$ becomes small — the two routes to surface limitation. The bulk layers are condensed into the permeances

$$\alpha = \frac{D_{\text{ox}} K_{\text{ox}}}{L_{\text{ox}}}, \quad \beta = \frac{D_{\text{m}} K_s}{L_{\text{m}}}, \quad \text{both } [\text{mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1/2}], \quad (35)$$

with bulk resistances $R_{\text{oxide}} = 1/\alpha$ and $R_{\text{metal}} = 1/\beta$.

Change of oxide dissolution convention. Note a deliberate difference from Model A: there, dissociation happens (fast) at the oxide–metal interface, so the oxide carries molecular H_2 and obeys Henry’s law (Eq. (5), driving force ΔP). In Model B, dissociation happens at the *gas–oxide* surface (Eq. (30)), so it is atomic hydrogen that enters the oxide and all bulk driving forces are expressed in $\sqrt{P}$ units — the code docstrings accordingly call K_{ox} the “oxide Sieverts constant” $[\text{mol m}^{-3} \text{ Pa}^{-1/2}]$. This makes α and β dimensionally identical, which is what allows the permeance-weighted interface pressure of Eq. (39).

3.3 L1+L6: perfect metal + surface kinetics

With no oxide, the coverage sits directly on the metal, so the interface driving force is $\sqrt{P_{\text{int}}} = g(\theta)$ and the metal flux is

$$J_{\text{m}}(\theta) = \beta(g(\theta) - \sqrt{P_{\text{down}}}). \quad (36)$$

Steady state is the single-unknown closure

$$J_{\text{surf}}(\theta) = J_{\text{m}}(\theta) \implies \theta_{\text{ss}}, \quad J_{\text{ss}} = \beta(g(\theta_{\text{ss}}) - \sqrt{P_{\text{down}}}), \quad (37)$$

solved by Brent’s method on $\theta \in (0, 1)$. The physics question this level answers: at what (P_{up}, T) does control shift from dissociation ($R_{\text{surface}} > R_{\text{metal}}$; flux $\propto P_{\text{up}}$, slope 1.0) to bulk diffusion ($R_{\text{metal}} > R_{\text{surface}}$; Sieverts recovered, slope 0.5)? High k_{diss} must reproduce Eq. (3) exactly — the L1+L6 parity check in the notebook.

3.4 L2a+L6: perfect oxide + surface kinetics

The same structure with the oxide as the only bulk layer:

$$J_{\text{ox}}(\theta) = \alpha(g(\theta) - \sqrt{P_{\text{down}}}), \quad J_{\text{surf}}(\theta) = J_{\text{ox}}(\theta) \implies \theta_{\text{ss}}. \quad (38)$$

Resistances R_{surface} (Eq. (34)) and $R_{\text{oxide}} = 1/\alpha$ decide whether the film or its surface is limiting.

3.5 L2+L6: oxide + metal + surface kinetics

Analytic reduction to one unknown. Three fluxes must match: surface (Eq. (30)), oxide, and metal. Nominally the unknowns are θ and P_{int} , but equating the oxide and metal fluxes, $\alpha(g(\theta) - \sqrt{P_{\text{int}}}) = \beta(\sqrt{P_{\text{int}}} - \sqrt{P_{\text{down}}})$, is *linear* in $\sqrt{P_{\text{int}}}$ and solves analytically:

$$\sqrt{P_{\text{int}}}(\theta) = \frac{\alpha g(\theta) + \beta \sqrt{P_{\text{down}}}}{\alpha + \beta}. \quad (39)$$

The interface pressure is the permeance-weighted mean of the two boundary driving forces. Substituting back leaves a single residual in θ :

$$J_{\text{surf}}(\theta) = J_{\text{ox}}(\theta) = J_{\text{m}}(\theta), \quad J_{\text{ox}} = \alpha(g(\theta) - \sqrt{P_{\text{int}}}(\theta)), \quad J_{\text{m}} = \beta(\sqrt{P_{\text{int}}}(\theta) - \sqrt{P_{\text{down}}}), \quad (40)$$

solved with Brent's method on $f(\theta) = J_{\text{surf}} - J_{\text{ox}} = 0$. This reduction (rather than nested 2-D solving) is what keeps the higher combinations tractable.

Rate-limiting analysis. Three series resistances, $R_{\text{total}} = R_{\text{surface}} + 1/\alpha + 1/\beta$, give fractional contributions

$$f_i = \frac{R_i}{R_{\text{total}}}, \quad i \in \{\text{surface, oxide, metal}\}, \quad (41)$$

and the step with $f_i > 0.5$ is declared rate-limiting (otherwise *mixed*). The pressure-exponent signatures — surface-limited $J \propto P^{1.0}$, oxide- or metal-limited $J \propto P^{0.5}$ under Model B's $g(\theta)$ formulation — are used in the notebook's regime maps over $(P_{\text{up}}, T, k_{\text{diss}})$.

3.6 L3+L6: defective oxide + surface kinetics

Each oxide path from L3 becomes an independent L2+L6 problem with its own permeance,

$$\alpha_{\text{intact}} = \frac{D_{\text{ox}} K_{\text{ox}}}{L_{\text{ox}}}, \quad \alpha_{\text{crack}} = \frac{\alpha_{\text{intact}}}{\gamma}, \quad \alpha_{\text{gb}} = \delta_D \alpha_{\text{intact}}, \quad \alpha_{\text{pinhole}} = \infty, \quad (42)$$

(γ, δ_D as in Eqs. (13)–(14)). Each path is solved for its own coverage θ_i and interface pressure $P_{\text{int},i}$ via Eqs. (39)–(40); the pinhole path replaces the oxide kinetics by metal surface kinetics ($k_{\text{diss}}^{\text{m}}, K_{\text{eq}}^{\text{m}}$, so $\sqrt{P_{\text{int}}} = g_{\text{m}}(\theta)$) or, if none are supplied, by the Sieverts limit $\sqrt{P_{\text{int}}} = \sqrt{P_{\text{up}}}$. The total is the area-weighted sum over intact plus all defect types,

$$J_{\text{total}} = f_{\text{intact}} J_{\text{intact}} + \sum_i f_i J_i, \quad f_{\text{intact}} = 1 - \sum_i f_i, \quad (43)$$

with f_i the area fraction of defect type i [–].

3.7 L2+L4+L6: oxide + defective metal + surface kinetics

The L2+L6 solver is wrapped in a self-consistency loop over the metal diffusivity. The metal permeance becomes

$$\beta_{\text{eff}} = \frac{D_{\text{eff}} K_{\text{s}}}{L_{\text{m}}}, \quad (44)$$

with D_{eff} from the combined microstructure model (Eq. (22)). Because trap occupancy depends on the local concentration, D_{eff} is evaluated on the through-thickness profile

$$C(x) = K_{\text{s}} \left[\sqrt{P_{\text{int}}} - (\sqrt{P_{\text{int}}} - \sqrt{P_{\text{down}}}) \frac{x}{L_{\text{m}}} \right], \quad 0 \leq x \leq L_{\text{m}}, \quad (45)$$

averaged (arithmetic/harmonic/inlet/outlet), and iterated: solve Eq. (40) with the current β_{eff} , rebuild the profile, update D_{eff} , repeat until the relative change in D_{eff} falls below tolerance. Outputs include the modification factor $M = D_{\text{eff}}/D_{\text{lattice}}$ and the resistance fractions of Eq. (41) evaluated with β_{eff} .

3.8 L3+L4+L6: full model

The most complete configuration composes everything: each oxide path of Eq. (42) is solved as an L2+L4+L6 problem (surface kinetics $\rightarrow$ path-specific $\alpha \rightarrow$ defective metal with iterated D_{eff}), then area-weighted:

$$J_{\text{total}} = f_{\text{intact}} J_{\text{intact}}^{\text{L2+L4+L6}} + \sum_i f_i J_i^{\text{L2+L4+L6}}. \quad (46)$$

Each path carries its own θ_i , $P_{\text{int},i}$, and $D_{\text{eff},i}$; the system-level rate-limiting step is taken from the path that dominates the total flux (or reported as *mixed* if no path contributes more than 70%).

4 Parameter values and sources

Both models are driven by a single unified configuration, `calculations/config/model_config.py`: the metal is X40 NiCrAlTi 31/19 (Incoloy 802) with transport data from [Schmidt and Batfalsky \(1985\)](#), the oxide is Cr_2O_3 (“sample 4”, the undamaged-chromia limit) assembled from [Nemanič et al. \(2023\)](#); [Stöver et al. \(1986\)](#); [Chen et al. \(2011\)](#), surface kinetics are the 316-steel measurements of [Grant et al. \(1988\)](#), and the metal microstructure (grain size, trap spectrum) is parameterised for Hastelloy-N-class Ni alloys from [Zhu et al. \(2021\)](#); [Lu et al. \(2022\)](#); [Young and Scully \(1997\)](#). Tables 4–6 list every parameter with its default, the range explored in the sensitivity analyses (`data/sensitivity_parameters.csv`), the source, and why that range was chosen. Entries marked `PLACEHOLDER` have no literature measurement; their defaults are assumptions and their ranges are deliberately wide. Primary-source raw data are archived in the literature-database workbook `permeation_literature_database_v3.xlsx` in `data/`.

4.1 Bulk transport and operating conditions

Table 4: Bulk transport parameters and operating conditions (used by both models). Defaults from `model_config.py`; ranges from the sensitivity configuration.

Parameter	Default	Range	Source and range justification
<i>Metal (Incoloy 802) — D_{m}, K_{s} in Eqs. (1), (3)</i>			
D_{ref} D.ref	$4.285 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$	$[4.3 \times 10^{-11}, 4.3 \times 10^{-7}]$	Schmidt and Batfalsky (1985) : T_2 permeation, X40 NiCrAlTi 31/19, 750–950 °C, 5 mm plate. ± 2 decades covers alloy-to-alloy scatter across the Ni–Cr family.
E_D E.D	72 200 J mol $^{-1}$	[60, 80] kJ mol $^{-1}$	Schmidt and Batfalsky (1985) . Window brackets reported diffusion activation energies for austenitic Ni–Cr alloys.
$K_{\text{s,ref}}$ K.s.ref	0.05093 mol m $^{-3}$ Pa $^{-1/2}$	$[10^{-4}, 10^{-1}]$	Schmidt and Batfalsky (1985) . Three-decade window spans reported Sieverts constants for austenitic alloys.
H_{s} H.s	7 200 J mol $^{-1}$	[1, 50] kJ mol $^{-1}$	Schmidt and Batfalsky (1985) . Solution enthalpy is small and weakly constrained; wide window for endothermic dissolvers.
T_{ref} (metal)	1223 K	[600, 1400] K	Schmidt and Batfalsky (1985) measurement reference (950 °C).
<i>Oxide (Cr_2O_3 sample 4) — D_{ox}, K_{ox} in Eqs. (5), (6)</i>			

Parameter	Default	Range	Source and range justification
$D_{\text{ox,ref}}$ D_ox_ref	$7.80 \times 10^{-19} \text{ m}^2 \text{ s}^{-1}$	$[7.8 \times 10^{-21}, 7.8 \times 10^{-17}]$	Nemanić et al. (2023) (Φ_{ox}, D); Stöver et al. (1986) (Q_p , Fig. 7 sample Cr2); Chen et al. (2011) (E_D). Undamaged-chromia limit; ± 2 decades for oxide quality/porosity variation.
$E_{D,\text{ox}}$ E_D_ox	$70\,434 \text{ J mol}^{-1}$	$[69, 71] \text{ kJ mol}^{-1}$	Chen et al. (2011) DFT (1.24 eV), consistent with Stöver et al. (1986) . Tight range: theory and experiment agree.
$K_{\text{ox,ref}}$ K_ox_ref	$0.35417 \text{ mol m}^{-3} \text{ Pa}^{-1/2}$	$[0.08, 0.4]$	Nemanić et al. (2023) + Stöver et al. (1986) : derived as $\Phi_{\text{ox}}/D_{\text{ox}}$ at $T_{\text{ref}} = 673 \text{ K}$ (Sieverts form, §3.2).
$H_{\text{sol,ox}}$ H_sol_ox	$163\,566 \text{ J mol}^{-1}$	$[20, 170] \text{ kJ mol}^{-1}$	$Q_{p,\text{ox}} - E_{D,\text{ox}}$ from Stöver et al. (1986) and Nemanić et al. (2023) ; wide range because Q_p measurements for chromia scatter strongly.
T_{ref} (oxide)	673 K	$[600, 1400] \text{ K}$	Nemanić et al. (2023) measurement reference (400°C).
<i>Geometry and operating conditions — $L_m, L_{\text{ox}}, P_{\text{up}}, T$</i>			
L_m L_metal	10^{-3} m	$[5 \times 10^{-4}, 5 \times 10^{-3}]$	Model default (CONDITIONS): 1 mm plate; range spans practical component wall thicknesses.
L_{ox} thickness	$4.8 \times 10^{-8} \text{ m}$	$[10^{-10}, 10^{-5}]$	Model default: sample-4 nominal 48 nm scale; range spans native film to thick thermally grown scale.
P_{up} P_upstream	10^5 Pa	$[10^{-7}, 10^7]$	Model default (CONDITIONS); range covers vacuum-facing to pressurised service.
T T_operating	873 K	$[573, 1273] \text{ K}$	Model default (CONDITIONS , 600°C); range spans the validated windows of the metal and oxide datasets.

4.2 Oxide defects (L3) and metal microstructure (L4)

Table 5: Defect and microstructure parameters. Trap binding energies are stored in `model_config.py` as $\text{eV} \times F$ with $F = 96\,485 \text{ C mol}^{-1}$.

Parameter	Default	Range	Source and range justification
<i>Oxide defects (L3) — f_i, γ, δ_D in Eqs. (12)–(15)</i>			
f_{pinhole}	0.01	$[10^{-5}, 0.1]$	Model default (OXIDE_DEFECTS); range from trace porosity to heavily damaged scale.
f_{crack}	0.005	$[10^{-5}, 0.1]$	Model default (OXIDE_DEFECTS).
$f_{\text{gb,defect}}$	0.005	$[10^{-5}, 0.1]$	Model default (OXIDE_DEFECTS).
γ thickness_factor	0.1	$[0.01, 0.5]$	Model default: crack retains 10% of oxide thickness (Eq. (13)); range from near-through-crack to half-thickness.
δ_D diffusivity_factor	10	$[1, 100]$	Model default: oxide-GB fast path (Eq. (14)); bounded by no enhancement and the fast-path limit.
<i>Metal microstructure (L4) — d, δ, N_L in Eqs. (16)–(22)</i>			
d grain_size	10^{-4} m	$[10^{-5}, 5 \times 10^{-4}]$	Zhu et al. (2021) : Hastelloy-N equiaxed grains $50\text{--}150 \mu\text{m}$ (EBSD); default is the midpoint, range extends to fine-grained condition.

Parameter	Default	Range	Source and range justification
δ <code>gb_thickness</code>	5×10^{-10} m	$[3 \times 10^{-10}, 10^{-9}]$	Nominal high-angle boundary width; no Hastelloy-N measurement available.
N_L <code>lattice_density</code>	8.774×10^{28} m $^{-3}$	$[8 \times 10^{28}, 1.2 \times 10^{29}]$	Fe BCC estimate (model default); range spans BCC to FCC interstitial site densities.
<i>Trap spectrum (L4) — $E_{b,i}$, $N_{T,i}$ in Eqs. (20)–(21)</i>			
E_b dislocations	19 297 J mol $^{-1}$ (0.20 eV)	[15, 25] kJ mol $^{-1}$	Lu et al. (2022): Alloy 625 TDS Peak 1, 0.186–0.215 eV (dislocation elastic field); default is the mean.
N_T dislocations	8.16×10^{12} m $^{-3}$	$[8.2 \times 10^{10}, 8.2 \times 10^{14}]$	Zhu et al. (2021): GND density from EBSD; ± 2 decades covers annealed to cold-worked states.
E_b grain boundaries	26 051 J mol $^{-1}$ (0.27 eV)	[25, 30] kJ mol $^{-1}$	Lu et al. (2022): TDS Peak 2, 0.258–0.281 eV (GB + carbide interface).
N_T grain boundaries	6×10^{14} m $^{-3}$	$[6 \times 10^{12}, 6 \times 10^{16}]$	Geometric estimate $3\rho_{gb}/d$ for the 100 μ m equiaxed grain (Eq. (16)); ± 2 decades tracks the grain-size range.
E_b vacancies	41 489 J mol $^{-1}$ (0.43 eV)	[35, 50] kJ mol $^{-1}$	Ni estimate; no Hastelloy-N vacancy trap measurement.
N_T vacancies	10^{26} m $^{-3}$	$[10^{24}, 10^{28}]$	Model default; no measurement — deliberately wide.
E_b carbides (M ₆ C)	26 051 J mol $^{-1}$ (0.27 eV)	[20, 40] kJ mol $^{-1}$	Lu et al. (2022): M ₆ C TDS, 0.258–0.281 eV; upper range allows for incoherent interfaces.
N_T carbides	2×10^{25} m $^{-3}$	$[2 \times 10^{23}, 2 \times 10^{27}]$	Young and Scully (1997): carbide trap density upper bound in Ni-17Cr-8Fe (TDS).

4.3 Surface kinetics (L6)

Table 6: Surface-kinetics parameters (Model B). Both parameter sets derive from Grant et al. (1988), the only literature source reporting the full dissociation constant k_1 ; the oxidised-foil condition proxies the Cr₂O₃ surface and the clean condition the bare metal (pinhole paths).

Parameter	Default	Range	Source and range justification
<i>Oxide surface (gas–Cr₂O₃) — k_{diss}, K_{eq} in Eqs. (28)–(30)</i>			
$k_{\text{diss,ref}}$ <code>k_diss_ref</code>	9.487×10^{-8} mol m $^{-2}$ s $^{-1}$ Pa $^{-1}$	$[9.5 \times 10^{-10}, 9.5 \times 10^{-6}]$	Grant et al. (1988): oxidised 316 foil, $k_1 = 9.2 \times 10^{-8}$ at 965 K, re-referenced to $T_{\text{ref}} = 1623$ K; ± 2 decades because 316 oxide is a proxy for Cr ₂ O ₃ .
E_{diss} <code>E_diss</code>	57 950 J mol $^{-1}$	[40, 70] kJ mol $^{-1}$	Grant et al. (1988): oxidised-both-surfaces condition ($E_{k_1} = 59 861$ J mol $^{-1}$; default shifted slightly by the re-referencing).

Parameter	Default	Range	Source and range justification
$K_{\text{eq,ref}}$ K_eq_ref	10^{-4} Pa^{-1}	$[10^{-14}, 10^{-2}]$	PLACEHOLDER: no gas–Cr ₂ O ₃ surface equilibrium constant in the literature; deliberately wide window bracketing the working default (± 2 decades above, extending down to the historical sweep floor).
H_{eq} H_eq	$20\,000 \text{ J mol}^{-1}$	$[5, 50] \text{ kJ mol}^{-1}$	PLACEHOLDER: no measured surface equilibrium enthalpy for Cr ₂ O ₃ .
T_{ref} (oxide surface)	1623 K	$[600, 1800] \text{ K}$	Configuration choice: matches the Cr ₂ O ₃ transport reference temperature.
<i>Metal surface (pinhole paths, L1+L6) — $k_{\text{diss}}^{\text{m}}, K_{\text{eq}}^{\text{m}}$</i> $k_{\text{diss,ref}}^{\text{m}}$ k_diss_metal_ref	$1.346 \times 10^{-6} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$	$[1.3 \times 10^{-8}, 1.3 \times 10^{-4}]$	Grant et al. (1988) : clean 316 surface, k_1 at 965 K; ± 2 decades for surface-condition uncertainty.
$E_{\text{diss}}^{\text{m}}$ E_diss_metal	$81\,560 \text{ J mol}^{-1}$	$[70, 100] \text{ kJ mol}^{-1}$	Grant et al. (1988) : clean-surface activation energy.
$K_{\text{eq,ref}}^{\text{m}}$ K_eq_metal_ref	10^{-3} Pa^{-1}	$[10^{-10}, 10^{-3}]$	PLACEHOLDER: no direct surface equilibrium measurement for Incoloy 802.
H_{eq}^{m} H_eq_metal	$15\,000 \text{ J mol}^{-1}$	$[5, 40] \text{ kJ mol}^{-1}$	PLACEHOLDER: no direct measurement.
T_{ref} (metal surface)	965 K	$[600, 1100] \text{ K}$	Grant et al. (1988) experimental reference temperature (692 °C).

5 Cross-model equation map

Table 7: Which equations are shared between the two models and which are specific. Shared equations are implemented once and imported by Model B.

Equation	Model A	Model B	Role in Model B
Sieverts’ law (1)	✓	✓	metal boundary condition via $g(\theta)$
Fick’s law (2)	✓	✓	unchanged bulk transport
Henry’s law (5)	✓	replaced	oxide carries atomic H — Sieverts-form $K_{\text{ox}}, \sqrt{P}$ driving force (§3.2)
Arrhenius form (26)	✓	✓	also applied to $k_{\text{diss}}, K_{\text{eq}}$
Parallel paths (12)	✓	✓	per-path L2+L6 solves (43)
Oriani trapping (21)	✓	✓	inside β_{eff} iteration (44)
GB enhancement (18)	✓	✓	inside β_{eff} iteration
Interface closure (8)	✓	replaced	analytic $\sqrt{P_{\text{int}}}(\theta)$ (39)
Regime ratio χ (10)	✓	generalised	three-way fractions (41)
Surface kinetics (28)–(34)	—	✓	the new series element
$g(\theta)$ conversion (32)	—	✓	reduces system to 1-D root find

The essential difference is the closure variable: Model A solves for the interface pressure P_{int} (Brent’s method on Eq. (8)/(25)), while Model B solves for the surface coverage θ (Brent’s method on Eq. (40)), with P_{int} recovered analytically from Eq. (39). In the fast-surface limit ($k_{\text{diss}} \rightarrow \infty$, $\theta \rightarrow \theta_{\text{eq}}$) every Model B level reduces to its Model A counterpart, which is the consistency test used throughout `Application/Surface_proposal.ipynb`.

References

- P. L. Andrew and A. A. Haasz. Models for hydrogen permeation in metals. *Journal of Applied Physics*, 72(7):2749–2757, 1992. doi: 10.1063/1.351526.
- M. I. Baskes. A calculation of the surface recombination rate constant for hydrogen isotopes on metals. *Journal of Nuclear Materials*, 92(2–3):318–324, 1980. doi: 10.1016/0022-3115(80)90117-8.
- A. M. Brass and A. Chanfreau. Accelerated diffusion of hydrogen along grain boundaries in nickel. *Acta Materialia*, 44(9):3823–3831, 1996. doi: 10.1016/1359-6454(95)00446-7.
- C. F. Chen, H. B. Yu, and S. Q. Zheng. First-principles study of hydrogen diffusion mechanism in Cr_2O_3 . *Science China Technological Sciences*, 54(1):88–94, 2011. doi: 10.1007/s11431-010-4112-3.
- Adolf Fick. Über diffusion. *Annalen der Physik*, 170(1):59–86, 1855. doi: 10.1002/andp.18551700105.
- D. M. Grant, D. L. Cummings, and D. A. Blackburn. Hydrogen in 316 steel — diffusion, permeation and surface reaction. *Journal of Nuclear Materials*, 152(2–3):139–145, 1988. doi: 10.1016/0022-3115(88)90128-7.
- E. W. Hart. On the role of dislocations in bulk diffusion. *Acta Metallurgica*, 5(10):597, 1957. doi: 10.1016/0001-6160(57)90127-X.
- G. W. Hollenberg, E. P. Simonen, G. Kalinin, and A. Terlain. Tritium/hydrogen barrier development. *Fusion Engineering and Design*, 28:190–208, 1995. doi: 10.1016/0920-3796(95)90039-X.
- Yaakov Kraftmakher. Equilibrium vacancies and thermophysical properties of metals. *Physics Reports*, 299(2–3):79–188, 1998. doi: 10.1016/S0370-1573(97)00082-3.
- Irving Langmuir. The adsorption of gases on plane surfaces of glass, mica and platinum. *Journal of the American Chemical Society*, 40(9):1361–1403, 1918. doi: 10.1021/ja02242a004.
- X. Lu, T. Depover, and R. Johnsen. Evaluation of hydrogen diffusion and trapping in nickel Alloy 625 by thermal desorption spectroscopy. *International Journal of Hydrogen Energy*, 47:31673–31683, 2022. doi: 10.1016/j.ijhydene.2022.07.094.
- A. McNabb and P. K. Foster. A new analysis of the diffusion of hydrogen in iron and ferritic steels. *Transactions of the Metallurgical Society of AIME*, 227:618–627, 1963.
- V. Nemanič, J. Kovač, M. Žumer, and J. Zavašnik. Impact of surface oxide on hydrogen permeability of chromium membranes. *International Journal of Hydrogen Energy*, 48:9723–9733, 2023. doi: 10.1016/j.ijhydene.2022.11.267.
- R. A. Oriani. The diffusion and trapping of hydrogen in steel. *Acta Metallurgica*, 18(1):147–157, 1970. doi: 10.1016/0001-6160(70)90078-7.
- A. Oudriss, J. Creus, J. Bouhattate, E. Conforto, C. Berziou, C. Savall, and X. Feaugas. Grain size and grain-boundary effects on diffusion and trapping of hydrogen in pure nickel. *Acta Materialia*, 60(19):6814–6828, 2012. doi: 10.1016/j.actamat.2012.09.004.

- G. Palumbo and K. T. Aust. Structure-dependence of intergranular corrosion in high purity nickel. *Acta Metallurgica et Materialia*, 38(11):2343–2352, 1990. doi: 10.1016/0956-7151(90)90101-L.
- M. A. Pick and K. Sonnenberg. A model for atomic hydrogen–metal interactions — application to recycling, recombination and permeation. *Journal of Nuclear Materials*, 131(2–3):208–220, 1985. doi: 10.1016/0022-3115(85)90459-3.
- O. W. Richardson, J. Nicol, and T. Parnell. The diffusion of hydrogen through hot platinum. *Philosophical Magazine, Series 6*, 8(43):1–29, 1904. doi: 10.1080/14786440409463168.
- H. Schmidt and P. Batfalsky. Hydrogen and tritium permeation through heat-resistant alloys. *Journal of Nuclear Materials*, 131, 1985. Source of the X40 NiCrAlTi 31/19 (Incoloy 802) transport parameters; H₂ and T₂ permeation in 30+ heat-resistant alloys.
- Adolf Sieverts. Die aufnahme von gasen durch metalle. *Zeitschrift für Metallkunde*, 21:37–46, 1929.
- D. Stöver, H. P. Buchkremer, and R. Hecker. Hydrogen and deuterium permeation through metallic and surface-oxidized chromium. *Surface and Coatings Technology*, 28:281–290, 1986. doi: 10.1016/0257-8972(86)90085-X.
- R. A. Strehlow and H. C. Savage. The permeation of hydrogen isotopes through structural metals at low pressures and through metals with oxide film barriers. *Nuclear Technology*, 22(1):127–137, 1974. doi: 10.13182/NT74-A31383.
- T. Tsuru and R. M. Latanision. Grain boundary transport of hydrogen in nickel. *Scripta Metallurgica*, 16(5):575–578, 1982. doi: 10.1016/0036-9748(82)90273-3.
- E. E. Underwood. *Quantitative Stereology*. Addison-Wesley, Reading, MA, 1970.
- W. R. Wampler. Surface recombination of hydrogen on clean nickel. *Applied Physics Letters*, 48(6):405–407, 1986. doi: 10.1063/1.96521.
- G. A. Young and J. R. Scully. Evidence that carbide precipitation produces hydrogen traps in Ni-17Cr-8Fe alloys. *Scripta Materialia*, 36(6):713–719, 1997. doi: 10.1016/S1359-6462(96)00440-X.
- A. S. Zarchy and R. C. Axtmann. Tritium permeation through 304 stainless steel at ultra-low pressures. *Journal of Nuclear Materials*, 79(1):110–117, 1979. doi: 10.1016/0022-3115(79)90436-4.
- H. Zhu, O. Muránsky, T. Wei, J. Davis, E. Budzakoska-Testone, H. Huang, and M. Drew. The effect of applied stress on the high-temperature creep behaviour and microstructure of NiMoCr Hastelloy-N alloy. *Materialia*, 16:101069, 2021. doi: 10.1016/j.mtla.2021.101069.

A Resistance formulation: complete derivation

This appendix derives the resistance quantities used in §2 from first principles and explains precisely in what sense the metal and oxide resistances can — and cannot — be compared. The corresponding code lives in `oxide_permeation.py` (functions `calculate_oxide_resistance`, `calculate_metal_resistance`, and `calculate_transition_pressure`) and in `interface_solver.py` (`calculate_oxide_metal_system`).

A.1 What a permeation resistance is

The electrical analogy defines a resistance through a linear response law,

$$J = \frac{X_{\text{up}} - X_{\text{down}}}{R_X}, \quad (47)$$

where X is some *driving-force variable* evaluated at the two boundaries, in analogy with $I = V/R$. The definition is only useful when J is genuinely linear in ΔX , because only then is R_X a constant of the layer, independent of operating point — exactly as an ohmic resistor has an R independent of the applied voltage. The subtlety in this model is that the two layers are linear in *different* variables: the choice of X is not free, it is dictated by each layer’s solubility law.

A.2 Oxide: linear in ΔP

Combining Henry’s law (Eq. (5)) at both faces with Fick’s first law across the thickness:

$$J_{\text{ox}} = D_{\text{ox}} \frac{C_{\text{up}} - C_{\text{down}}}{L_{\text{ox}}} = D_{\text{ox}} \frac{K_{\text{ox}} P_{\text{up}} - K_{\text{ox}} P_{\text{down}}}{L_{\text{ox}}} = \underbrace{\frac{D_{\text{ox}} K_{\text{ox}}}{L_{\text{ox}}}}_{1/R_{\text{oxide}}} (P_{\text{up}} - P_{\text{down}}). \quad (48)$$

The flux is exactly linear in ΔP , so the oxide possesses a true, pressure-independent resistance

$$R_{\text{oxide}} = \frac{L_{\text{ox}}}{D_{\text{ox}} K_{\text{ox}}} \quad [\text{Pa s m}^2 \text{ mol}^{-1}], \quad (49)$$

with the pressure difference itself as the driving force ($X = P$). Every extra pascal of driving force buys the same extra flux, at any operating pressure.

A.3 Metal: linear in $\Delta\sqrt{P}$

The metal transports a different species. In the L2b picture the intact H_2 molecule arriving through the oxide dissociates at the oxide–metal interface,



and it is the *atoms* that dissolve into and diffuse through the metal lattice. The law of mass action for this equilibrium, with two atoms produced per molecule, reads $C_{\text{H}}^2/P_{\text{int}} = \text{const}$, so the dissolved-atom concentration scales as $C_{\text{H}} = K_{\text{s}}\sqrt{P_{\text{int}}}$ — Sieverts’ law (Eq. (1)) is nothing but the 2:1 stoichiometry of reaction (50). Repeating the resistance construction with this boundary condition:

$$J_{\text{m}} = D_{\text{m}} \frac{K_{\text{s}}\sqrt{P_{\text{up}}} - K_{\text{s}}\sqrt{P_{\text{down}}}}{L_{\text{m}}} = \underbrace{\frac{D_{\text{m}} K_{\text{s}}}{L_{\text{m}}}}_{1/R_{\text{L1}}} (\sqrt{P_{\text{up}}} - \sqrt{P_{\text{down}}}). \quad (51)$$

The metal is exactly linear too — but in $\Delta\sqrt{P}$, not ΔP . Its natural resistance is

$$R_{\text{L1}} = \frac{L_{\text{m}}}{D_{\text{m}} K_{\text{s}}} = \frac{L_{\text{m}}}{\Phi} \quad [\text{Pa}^{1/2} \text{ s m}^2 \text{ mol}^{-1}], \quad (52)$$

a constant of the layer, with driving force $X = \sqrt{P}$.

A.4 Why the two are not directly comparable

Both R_{oxide} and R_{L1} are legitimate constants, but they act on different driving forces and carry different units:

Layer	Flux law	Driving force X	Resistance	Units
Oxide	$J = \Delta P / R_{\text{oxide}}$	P	$R_{\text{oxide}} = L_{\text{ox}} / (D_{\text{ox}} K_{\text{ox}})$	$\text{Pa s m}^2 \text{ mol}^{-1}$
Metal	$J = \Delta\sqrt{P} / R_{\text{L1}}$	$\sqrt{P}$	$R_{\text{L1}} = L_{\text{m}} / (D_{\text{m}} K_{\text{s}})$	$\text{Pa}^{1/2} \text{ s m}^2 \text{ mol}^{-1}$

Asking whether $R_{\text{oxide}} > R_{\text{L1}}$ is dimensionally meaningless — like comparing a quantity in ohms with one in kilograms. Chemically, the mismatch is the dissociation step (50) sitting between the two layers: the oxide counts *molecules* (one dissolved per gas molecule, hence $\propto P$), the metal counts *atoms* produced by an equilibrium that delivers two per molecule (hence $\propto \sqrt{P}$). The unit $\text{Pa}^{1/2}$ separating the two resistances is exactly the stoichiometric square root introduced by that reaction. Yet the L2b regime question (“which layer limits the flux?”) demands a comparison. The resolution is to re-express the metal on the oxide’s per-pascal basis.

A.5 Linearising the metal: origin of the factor 2

Fix the downstream pressure and consider the metal flux as a function of its upstream (interface) pressure, $J_{\text{m}}(P) = \frac{D_{\text{m}}K_{\text{s}}}{L_{\text{m}}}(\sqrt{P} - \sqrt{P_{\text{down}}})$. Taylor-expanding about the operating point $P = P_{\text{int}}$:

$$J_{\text{m}}(P) \approx J_{\text{m}}(P_{\text{int}}) + \left. \frac{dJ_{\text{m}}}{dP} \right|_{P_{\text{int}}} (P - P_{\text{int}}), \quad \left. \frac{dJ_{\text{m}}}{dP} \right|_{P_{\text{int}}} = \frac{D_{\text{m}}K_{\text{s}}}{L_{\text{m}}} \cdot \left. \frac{d\sqrt{P}}{dP} \right|_{P_{\text{int}}} = \frac{D_{\text{m}}K_{\text{s}}}{L_{\text{m}}} \cdot \frac{1}{2\sqrt{P_{\text{int}}}}. \quad (53)$$

The local slope dJ/dP is a per-pascal conductance; its reciprocal is the metal’s *effective* per-pascal resistance at that operating point,

$$R_{\text{metal}}(P_{\text{int}}) = \left(\left. \frac{dJ_{\text{m}}}{dP} \right|_{P_{\text{int}}} \right)^{-1} = \frac{2 L_{\text{m}} \sqrt{P_{\text{int}}}}{D_{\text{m}} K_{\text{s}}} \quad [\text{Pa s m}^2 \text{ mol}^{-1}], \quad (54)$$

which is Eq. (9). The factor 2 is nothing but the derivative $d\sqrt{P}/dP = 1/(2\sqrt{P})$ — no additional physics. The approximation is valid when the pressure drop across the metal is small compared with P_{int} , i.e. near the operating point about which the expansion was taken. Physically: the dissociation equilibrium (50) yields dissolved atoms only as $\sqrt{P}$, so at higher pressure each additional pascal of H_2 converts into proportionally fewer new H^* atoms — the marginal atom yield falls as $1/(2\sqrt{P_{\text{int}}})$. Each additional pascal therefore buys less additional flux, and the metal’s per-pascal resistance *grows* as $\sqrt{P_{\text{int}}}$ while the oxide’s, which counts molecules one-for-one, stays constant.

A.6 Consequences: regime ratio and transition pressure

With both layers on the per-pascal basis, the ratio $\chi = R_{\text{oxide}}/R_{\text{metal}}$ of Eq. (10) is dimensionless and meaningful. Since $R_{\text{metal}} \propto \sqrt{P_{\text{int}}}$, χ *falls* as pressure rises: the bilayer is oxide-limited at low pressure (flux slope $\rightarrow 1.0$) and metal-limited at high pressure (slope $\rightarrow 0.5$). The crossover follows from setting $R_{\text{oxide}} = R_{\text{metal}}$:

$$\frac{L_{\text{ox}}}{D_{\text{ox}}K_{\text{ox}}} = \frac{2 L_{\text{m}} \sqrt{P_{\text{trans}}}}{D_{\text{m}}K_{\text{s}}} \implies \sqrt{P_{\text{trans}}} = \frac{L_{\text{ox}} D_{\text{m}}K_{\text{s}}}{2 D_{\text{ox}}K_{\text{ox}} L_{\text{m}}} \implies P_{\text{trans}} = \left(\frac{L_{\text{ox}} D_{\text{m}}K_{\text{s}}}{2 D_{\text{ox}}K_{\text{ox}} L_{\text{m}}} \right)^2, \quad (55)$$

which is Eq. (11).

A.7 Caveats and the Model B alternative

Resistances are diagnostics, not the solution. Because the series combination of a linear element (R_{oxide}) and a linearised one (R_{metal}) is only approximate, the L2b flux is *never* computed as $\Delta P/(R_{\text{oxide}} + R_{\text{metal}})$. The code always solves the exact flux-continuity condition Eq. (8) for P_{int} by Brent’s method and only afterwards evaluates R_{oxide} , $R_{\text{metal}}(P_{\text{int}})$, and χ to label the regime (`interface_solver.py` $\rightarrow$ `calculate_oxide_metal_system`).

Model B sidesteps the mismatch. In the surface-kinetics model every layer is expressed in $\sqrt{P}$ units from the outset: dissociation at the gas–oxide surface means atomic hydrogen traverses the oxide (§3.2), so the permeances α and β of Eq. (35) are dimensionally identical and the resistances R_{surface} , $1/\alpha$, $1/\beta$ add and compare directly (Eq. (41)) — no linearisation, and no factor of 2, is ever needed there.